REWARD CENTERING

Reinforcement Learning Conference
11 August 2024

Abhishek Naik

with thanks to Yi Wan, Manan Tomar, and Rich Sutton

amii UNIVERSITY OF ALBERTA RL & AI

THE MAIN MESSAGE

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$$Q_{t+1}(S_t, A_t) \doteq Q_t(S_t, A_t) + \alpha_t \left[R_{t+1} + \gamma \max_{a'} Q_t(S_{t+1}, a') - Q_t(S_t, A_t) \right]$$

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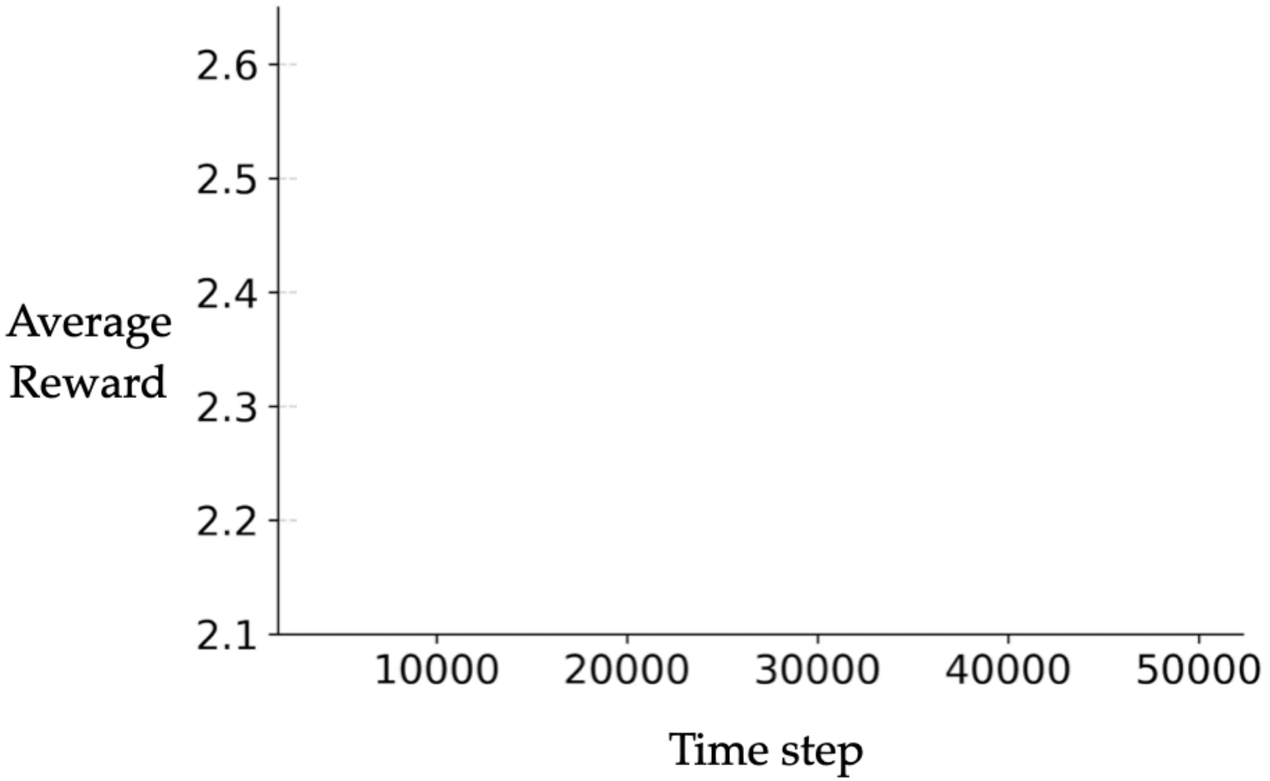
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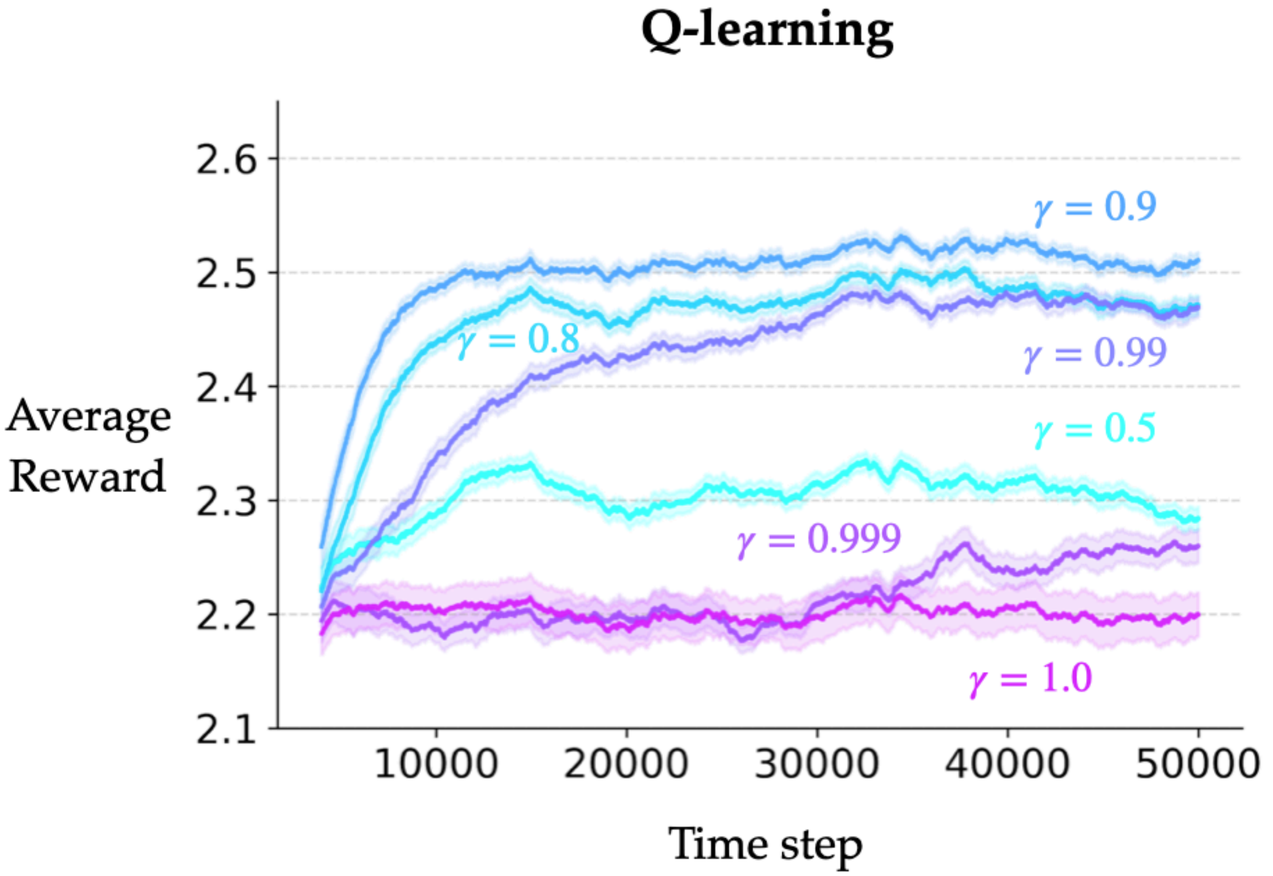
$$S_0\ A_0\ R_1\ S_1\ A_1, R_2 \ldots\ S_t\ A_t\ R_{t+1}\ S_{t+1}\ A_{t+1}\ R_{t+2}\ \ldots$$

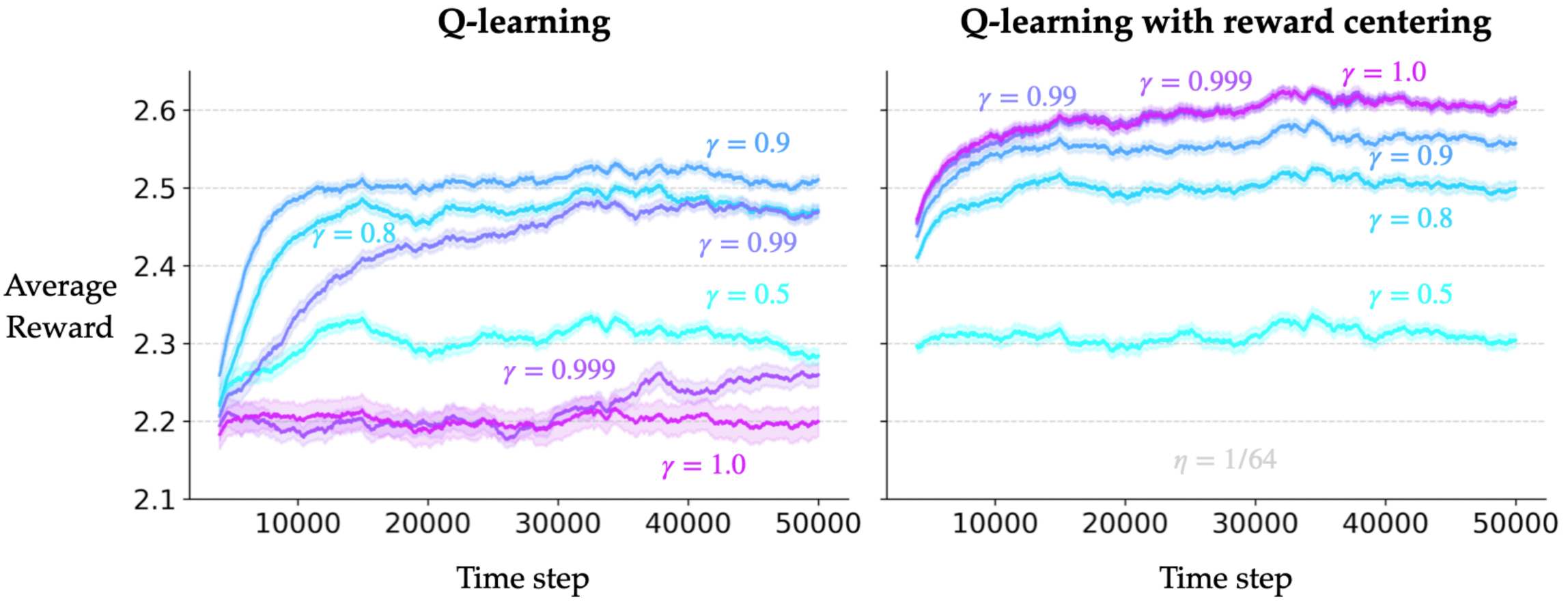
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$$\downarrow$$

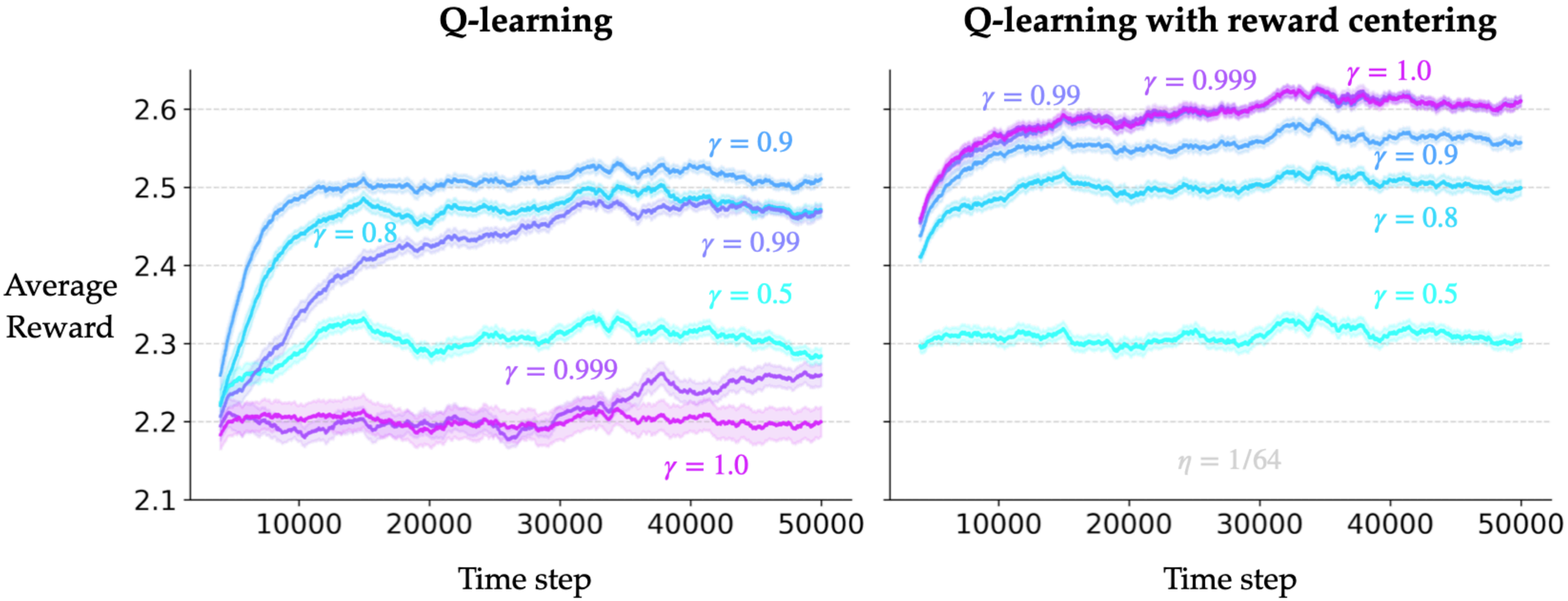
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Average Reward
2.6
2.5
2.4
2.3
2.2
2.1
10000
20000
30000
40000
50000
Time step

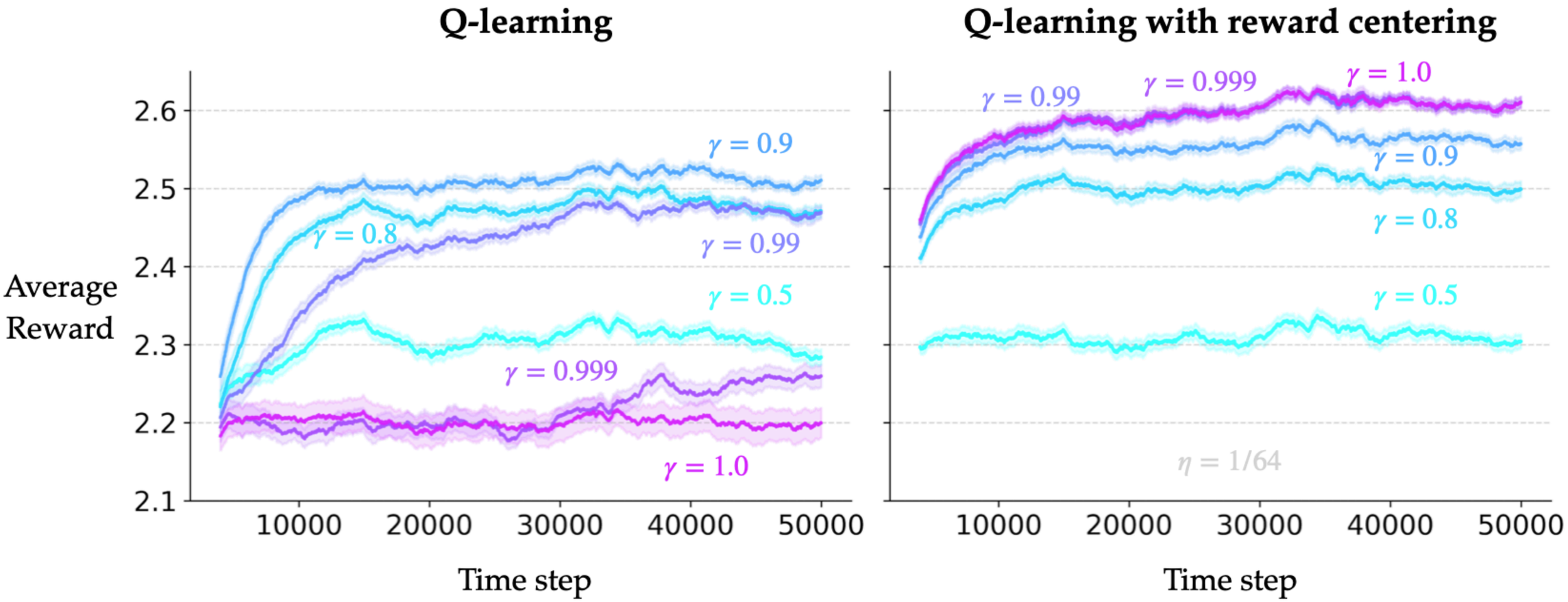

Q-learning
Average Reward
2.6
2.5
2.4
2.3
2.2
2.1
$\gamma = 0.9$
$\gamma = 0.8$
$\gamma = 0.99$
$\gamma = 0.5$
$\gamma = 0.999$
$\gamma = 1.0$
10000
20000
30000
40000
50000
Time step


Q-learning
Q-learning with reward centering
Average Reward
2.6
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$\eta = 1/64$
10000
20000
30000
40000
50000
Time step
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NO INSTABILITY WITH LARGE DISCOUNT FACTORS



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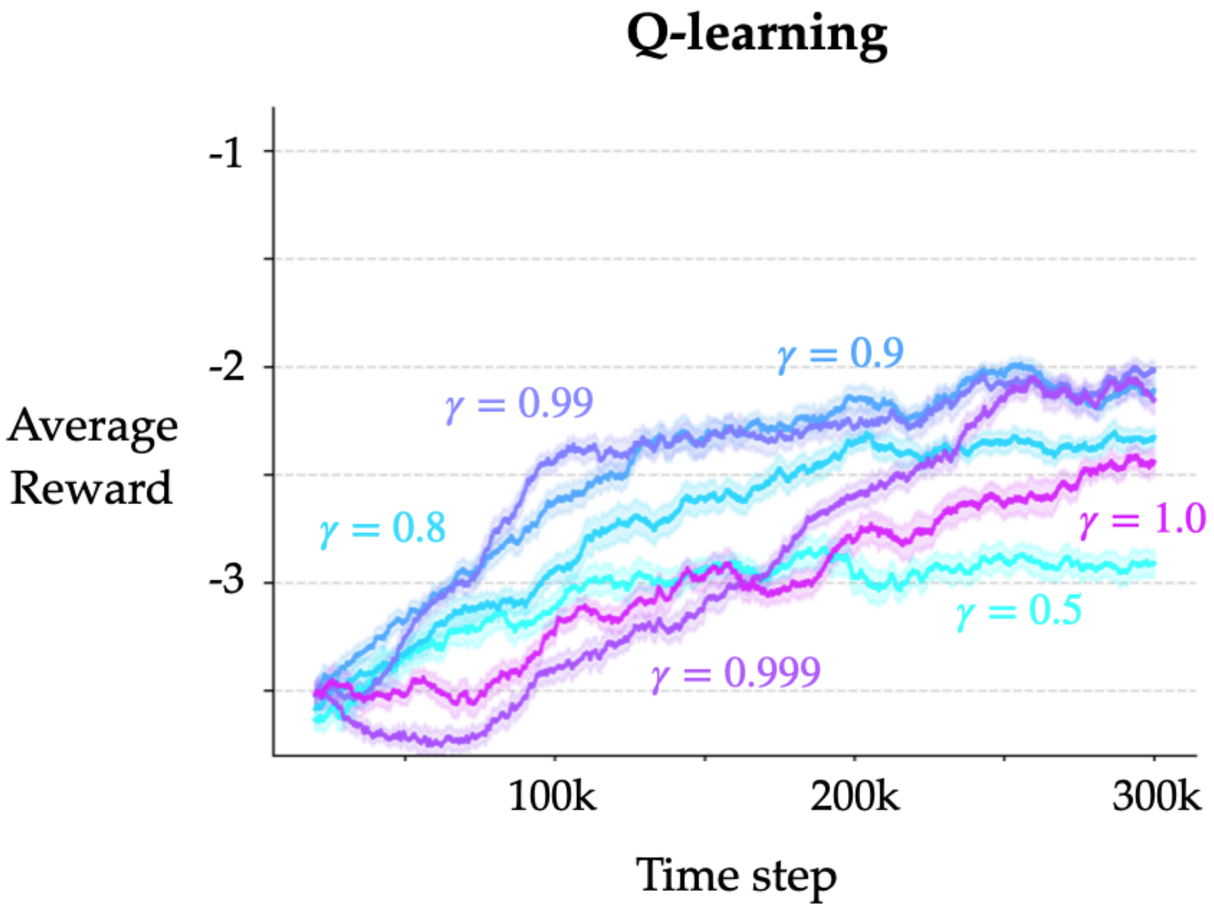


AccessControl (tabular)

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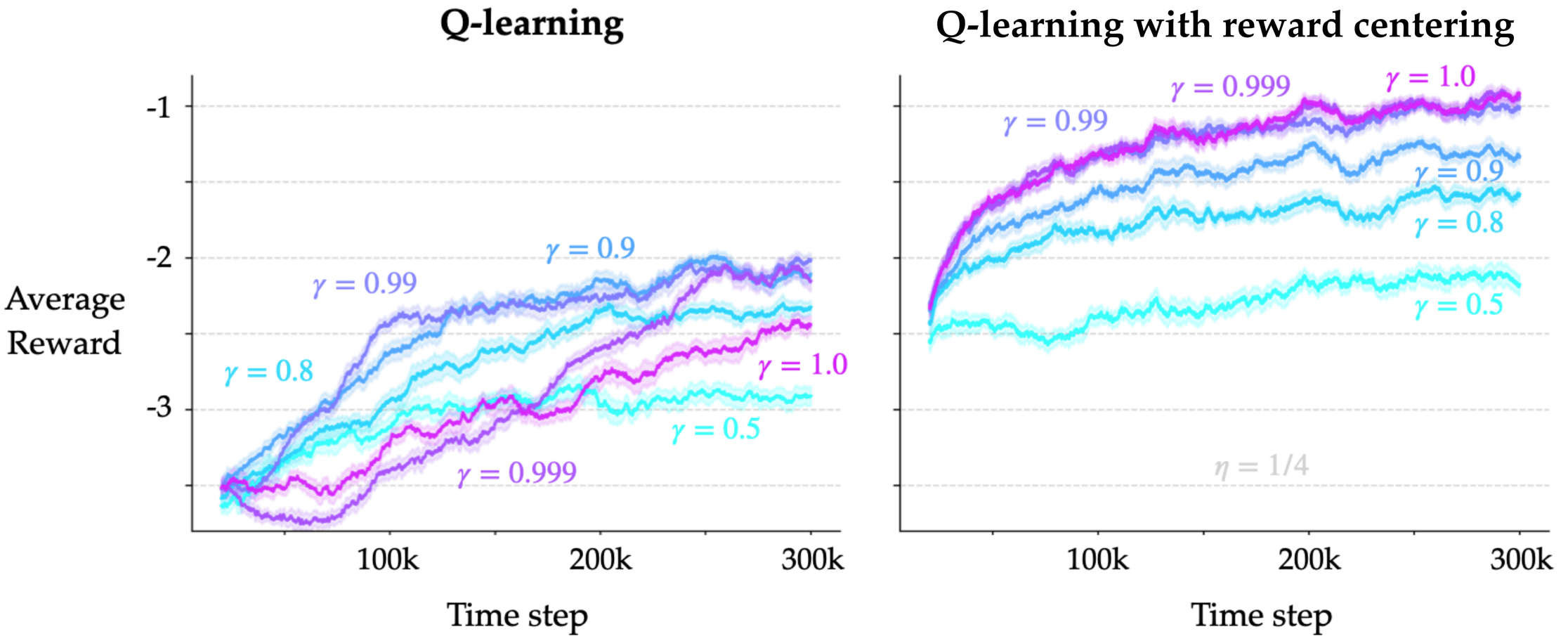
PuckWorld (linear FA)

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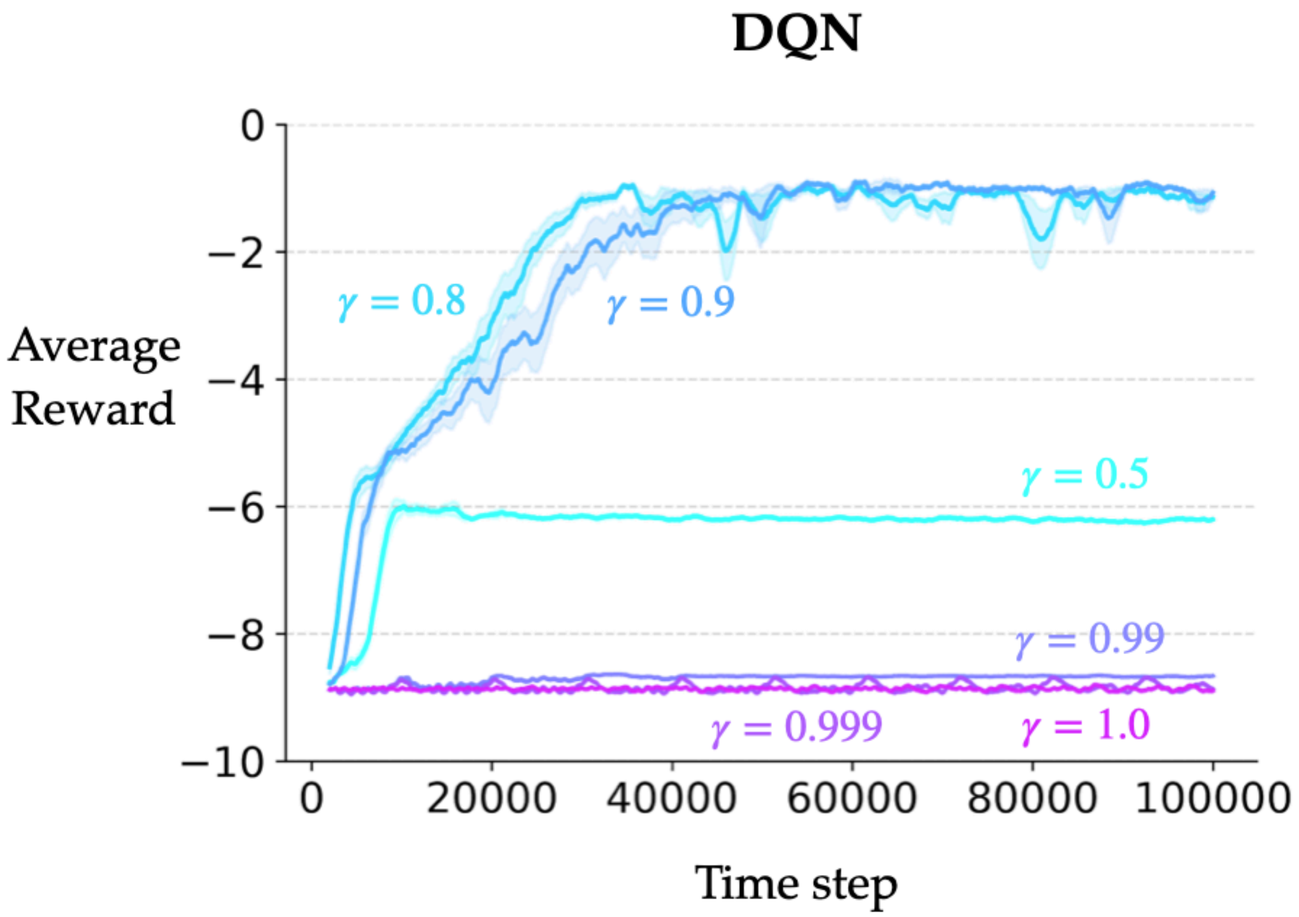


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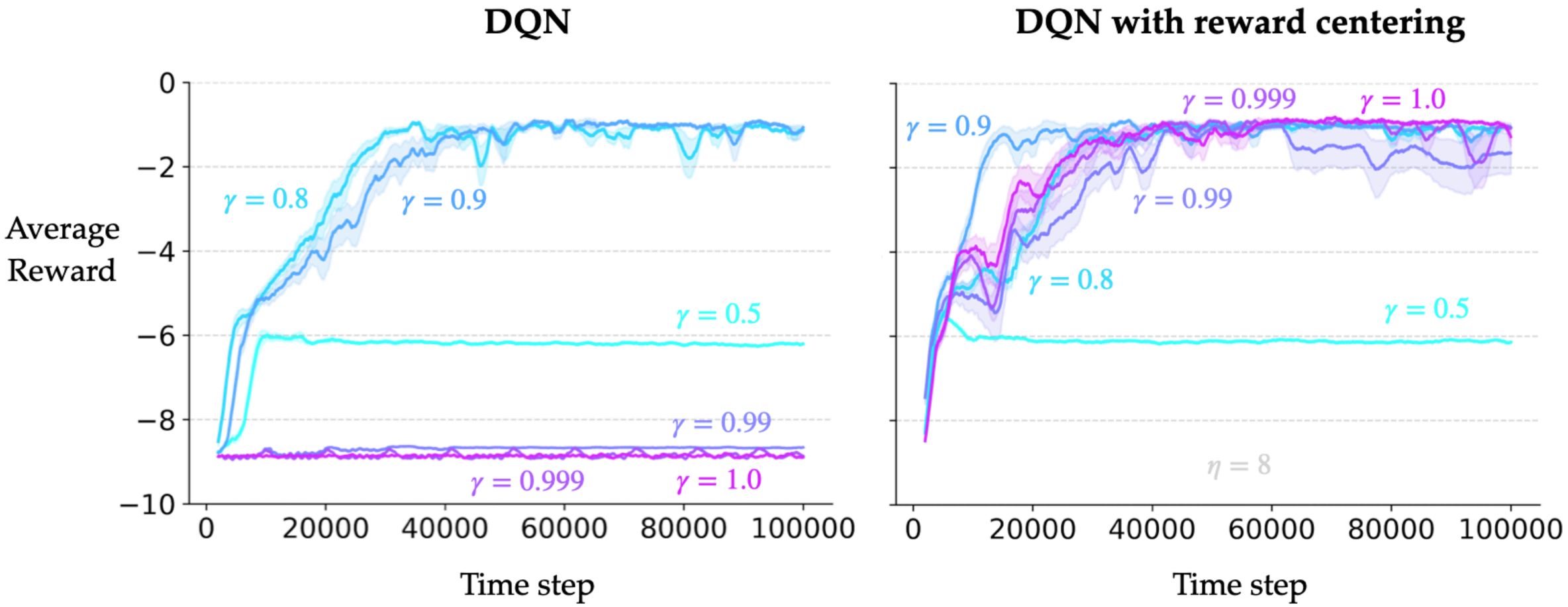
Pendulum (non-linear FA)

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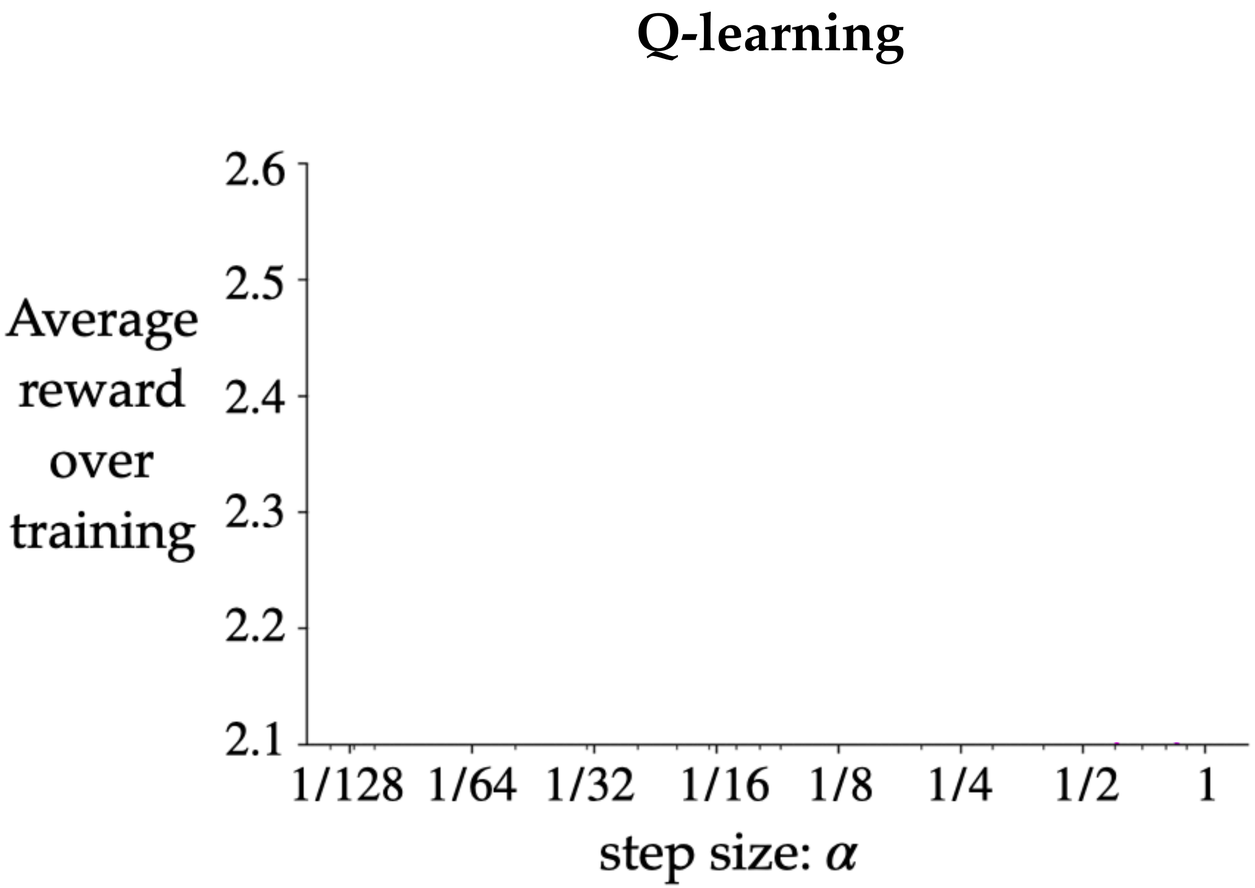
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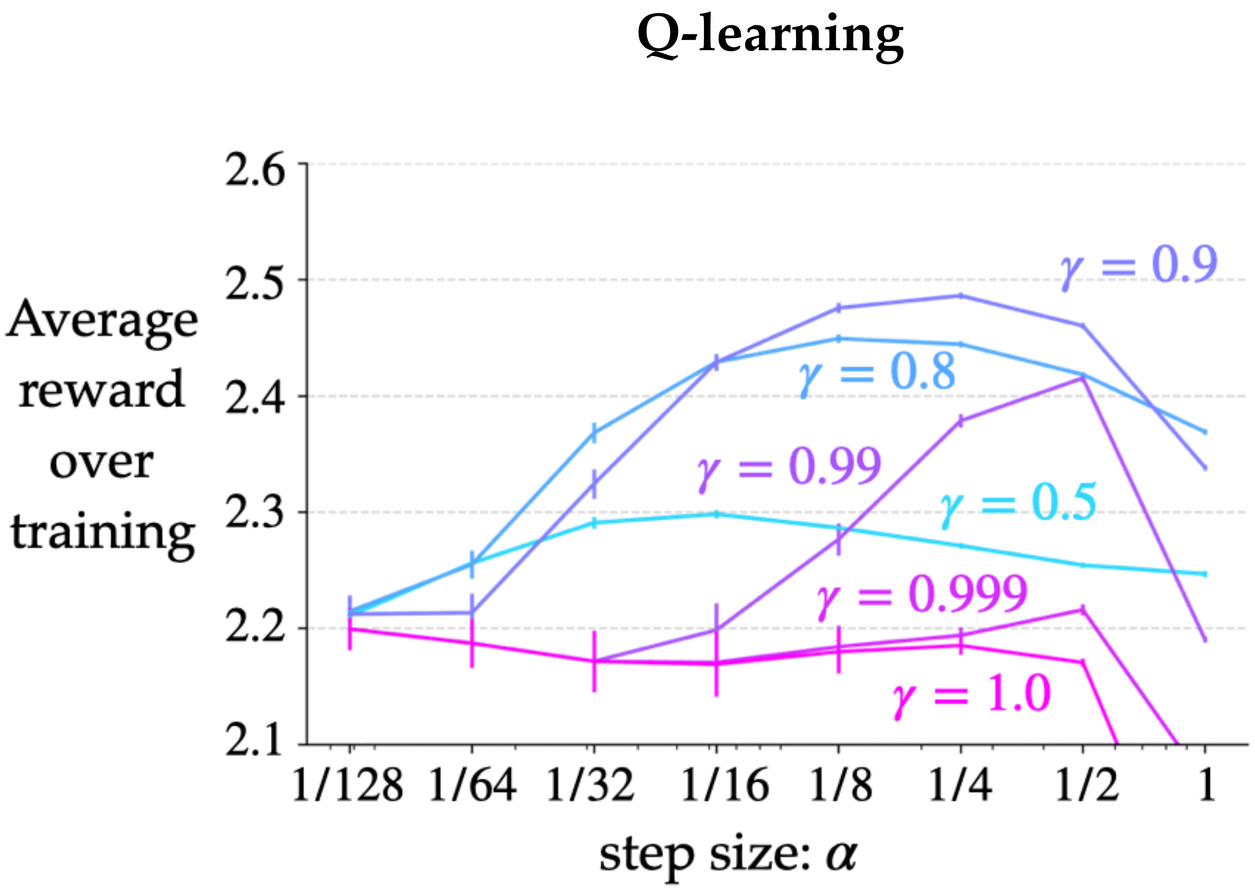


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TRENDS ARE CONSISTENT ACROSS PARAMETERS

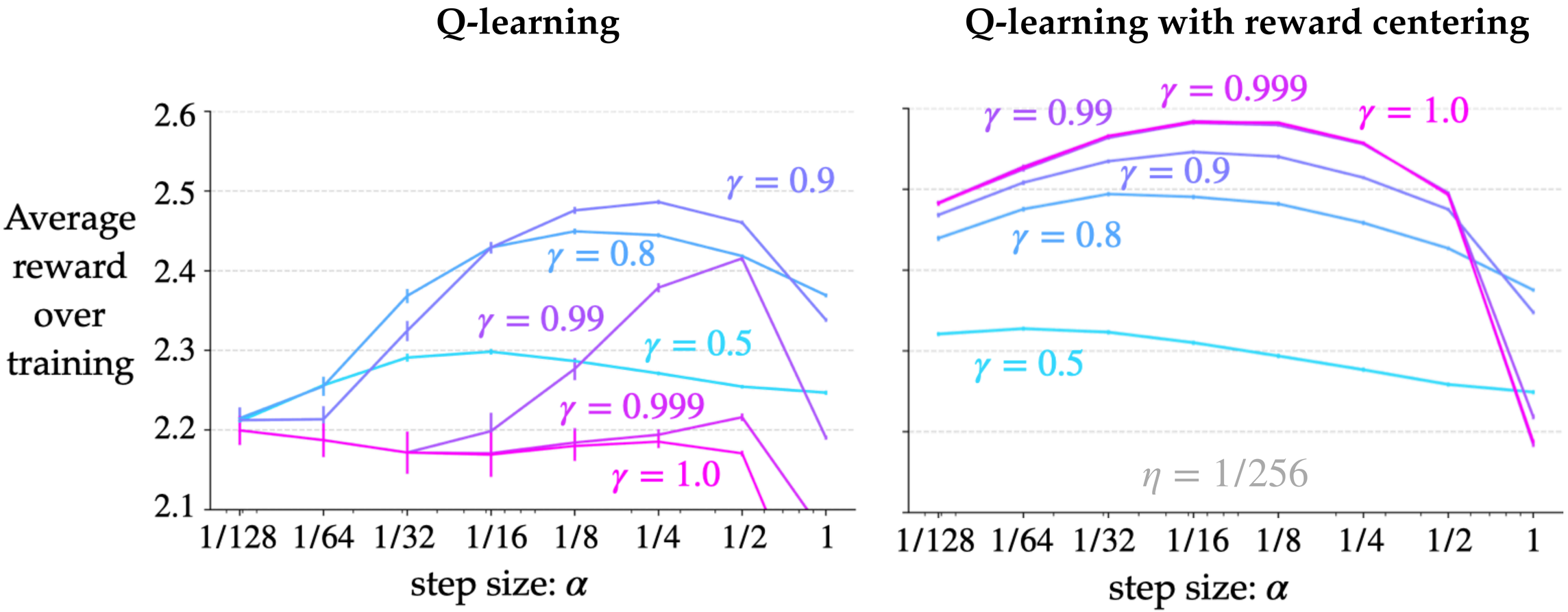


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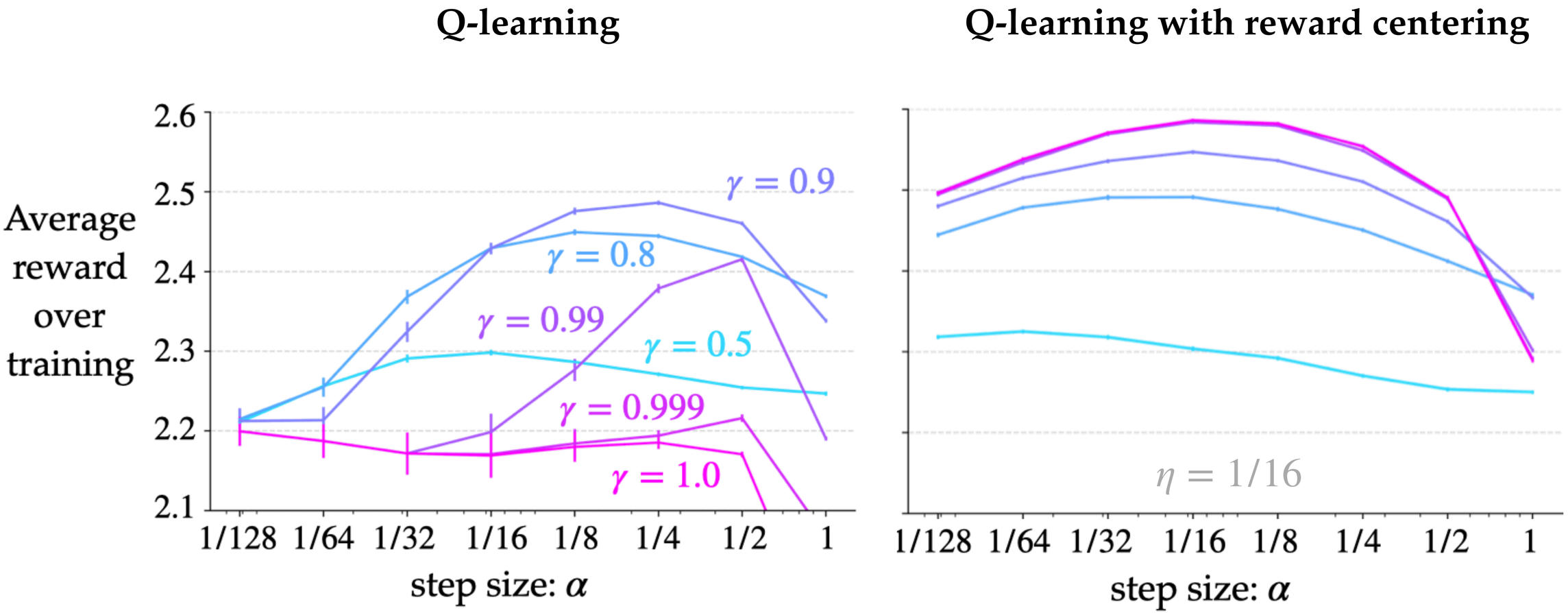
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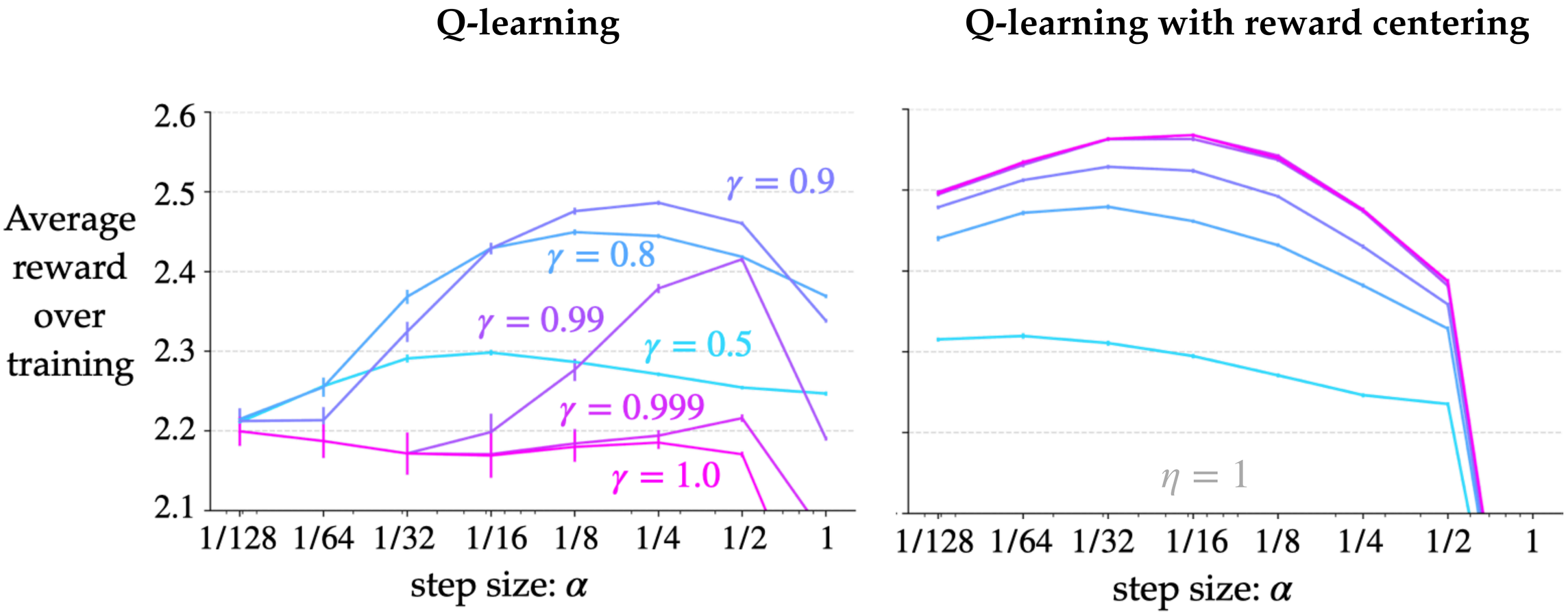
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Standard discounted value function

$$v_\pi^\gamma(s) \doteq \mathbb{E}_\pi\left[R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \ldots \,|\, S_t = s\right] = \mathbb{E}_\pi\left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \,|\, S_t = s\right]$$

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Average reward

$$r(\pi) \doteq \lim_{n\to\infty} \frac{1}{n} \mathbb{E}_\pi\left[\sum_{t=1}^{n} R_t\right]$$

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Differential value function

$$\tilde{v}_\pi(s) \doteq \mathbb{E}_\pi[R_{t+1} - r(\pi) + R_{t+2} - r(\pi) + \ldots \,|\, S_t = s]$$

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Differential value function

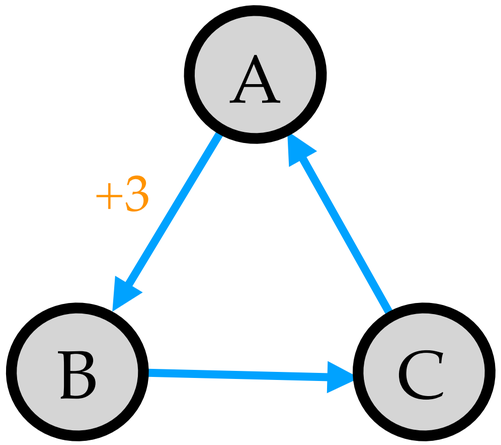
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INTUITION
THROUGH AN EXAMPLE

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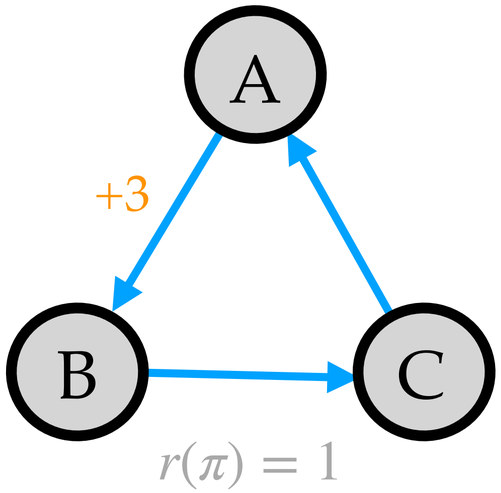
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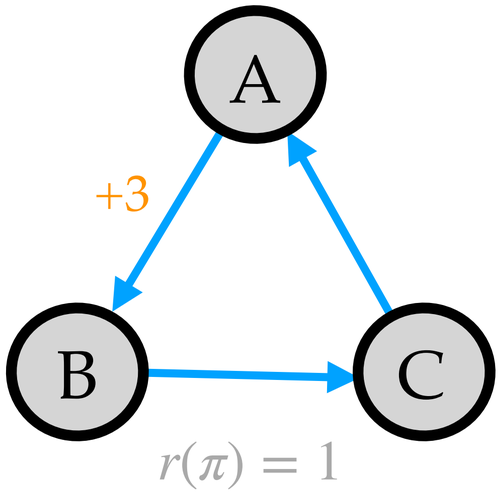
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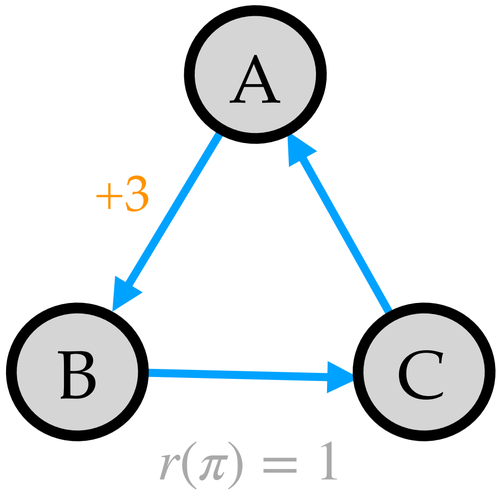
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	s_A	s_B	s_C
Differential values	1	-1	0

INTUITION THROUGH AN EXAMPLE

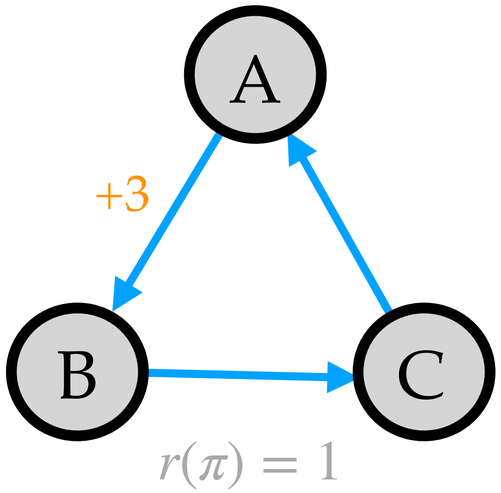
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Standard discounted values	$\gamma = 0.8$ $\gamma = 0.9$ $\gamma = 0.99$			
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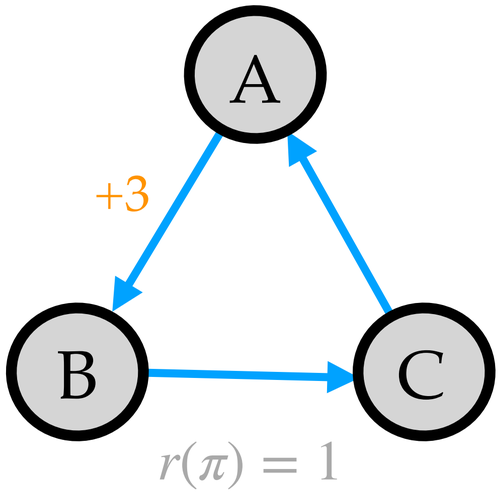
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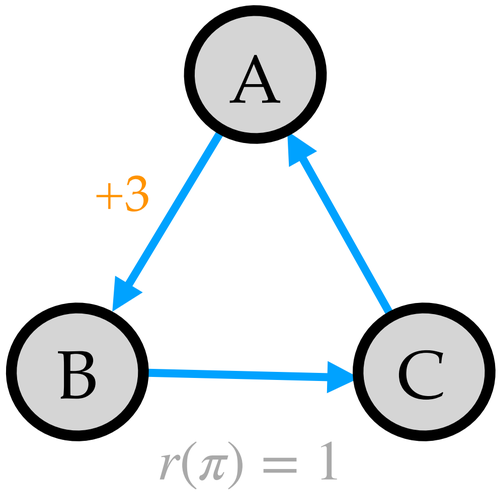
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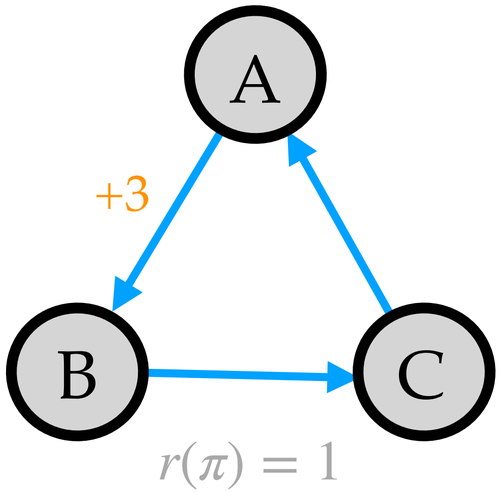
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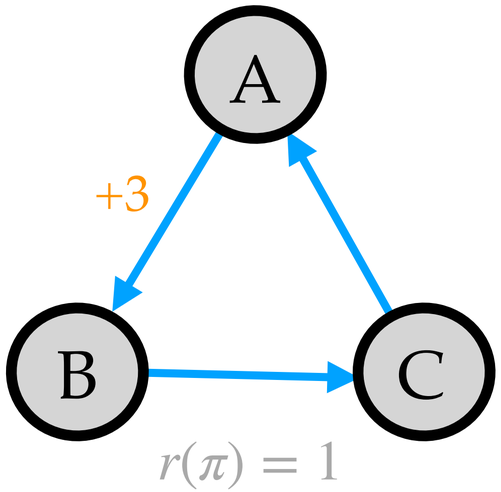
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		s_A	s_B	s_C	$\frac{r(\pi)}{1-\gamma}$
Standard discounted values	$\gamma = 0.8$	6.15	3.93	4.92	5
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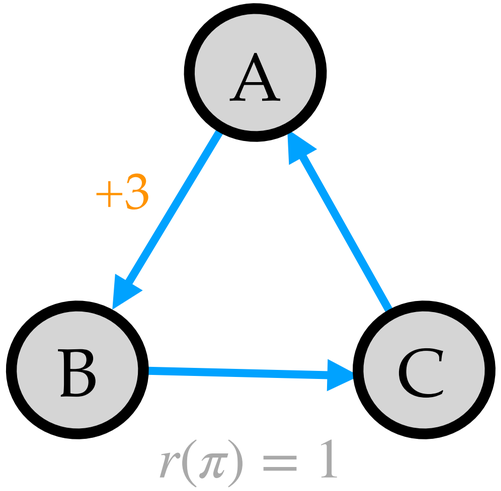
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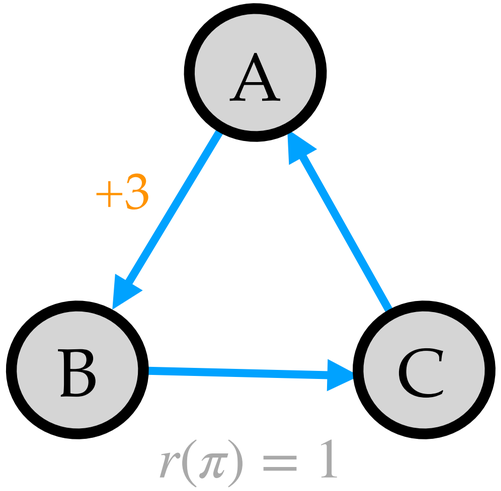
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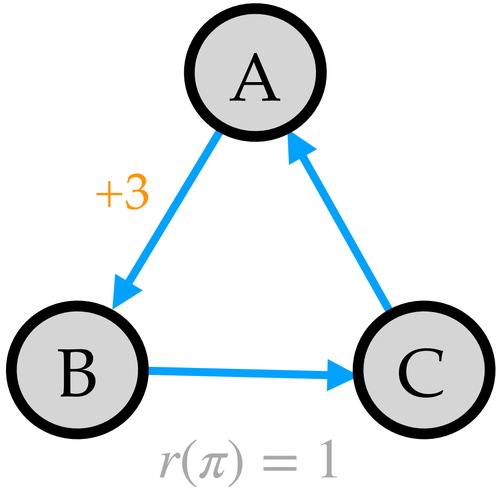
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	$\gamma = 0.99$	101.01	98.99	99.99	100
	$\gamma = 0.8$	1.15	-1.07	-0.08	
	$\gamma = 0.9$	1.07	-1.03	-0.04	
	$\gamma = 0.99$	1.01	-1.01	-0.01	
Differential values		1	-1	0	

INTUITION THROUGH AN EXAMPLE

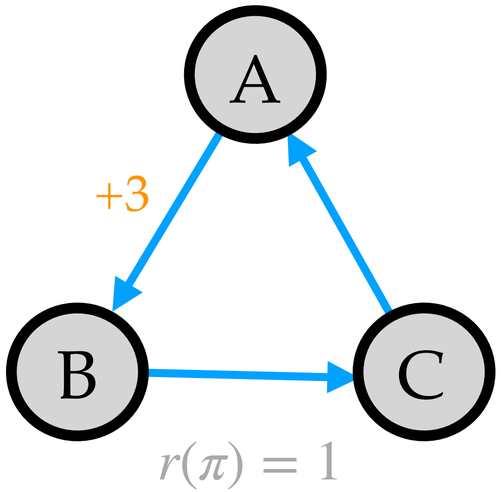
$$v_\pi^\gamma(s) = \frac{r(\pi)}{1-\gamma} + \tilde{v}_\pi(s) + e_\pi^\gamma(s)$$



		s_A	s_B	s_C	$\frac{r(\pi)}{1-\gamma}$
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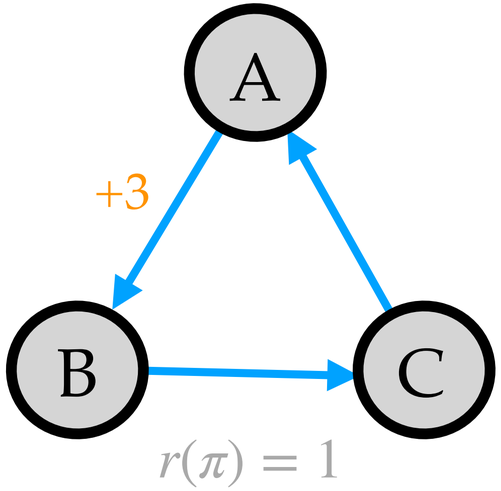
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Centered discounted values	$\gamma = 0.8$	1.15	-1.07	-0.08	
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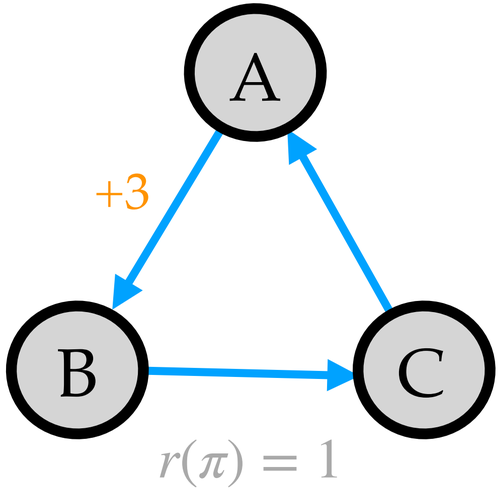
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Centered discounted value function

$$\tilde{v}_\pi^\gamma(s) \doteq \mathbb{E}_\pi\Big[\sum_{k=0}^{\infty} \gamma^k \big(R_{t+k+1} - r(\pi)\big) \,|\, S_t = s\Big]$$

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THANK YOU